

S. AMRITHA

Perungudi, Chennai 600096, India | +91 86676 14135 | amritha16112005@gmail.com

[in linkedin.com/in/amritha-s-oo](https://www.linkedin.com/in/amritha-s-oo) github.com/Amritha902 [Credly](https://www.credly.com)



PROFESSIONAL SUMMARY

Final-year Electronics and Communication Engineering student at VIT Chennai, reading a parallel BS in Data Science at IIT Madras. I work where networks meet the cloud: backbone topology and fault correlation, queue behaviour on Linux, and the Azure fabric layered over it, turning raw operational telemetry into decisions an on-call engineer can act on. I am equally interested in financial technology, where market data, risk, and regulatory compliance raise the same questions about trusting data at scale.

EXPERIENCE

Microsoft | Cloud Network Engineer Intern, Network Engineering Services (D&D) | Hyderabad May 2026 - Jul 2026

- Built Argus, a WAN fault-correlation engine that collapses ICM incident storms from one physical fault into a single root-cause group over a dual-graph topology model: 83.8% group recall, 93.1% purity, P99 under 15 s, shipped as an Azure Function App.
- Traced every missed incident pair to its cause, identified cross-site link adjacency as the dominant gap, and added a four-lever learning layer that produced the largest recall gains.

Centre of Nanoelectronics & VLSI, VIT Chennai | Summer Research Intern 2025

- Designed a delay-optimised low-power multiplier with input-based bypass logic in Cadence Virtuoso at 180 nm, the basis of the patent below.

RESEARCH & PATENTS

- Continuous Privacy-Preserving Compliance Verification Using Zero-Knowledge Proof Graphs.** Indian Patent App. No. [202641092083](#) (Co-Inventor), applicant VIT Chennai; filed 29 Jul 2026, published under Section 11A on 07 Aug 2026. Domain: applied cryptography and regulatory technology - zero-knowledge proofs, trusted execution, and recursive graph hashing to verify compliance continuously without exposing underlying data.
- Power-Gated Full Adder Cell with Input-Based Bypass Logic and Multiplier Architecture.** Indian Patent App. No. [202641062765](#) (Co-Inventor), published. Domain: low-power VLSI arithmetic - NOR-based power gating with input-driven bypass logic cutting switching and leakage power at 180 nm, extended to a full multiplier.

TECHNICAL PROJECTS

- ACAPE** - Adaptive queue management: fq_codel tuning driven by eBPF telemetry with a three-timescale controller, trading queue delay against throughput on Linux tc qdiscs.
- SERUM** - Content-aware containment framework for agentic malware, detecting by behaviour rather than signature matching.
- RFCGuardian** - Temporal validity in retrieval over the versioned IETF RFC corpus: 25% of BM25 queries return obsoleted normative text at rank 1; built a validity-aware ranker at MRR 0.773 with harmful-at-1 down 40%.
- InfoCrux** - AI market credibility engine; Winner, Jury Recommendation Prize at the SEBI Securities Market Hackathon 2025, top 6 nationally. Streamlit platform with 0-100 credibility scoring, NSE/BSE APIs, VirusTotal and URLScan checks, and CrewAI workflows on Gemini.
- VidyaVahini** - Offline-first AI education PWA (React, FastAPI, Firebase, Vertex AI) with auth, roles, and analytics; Top 60 at the Google Generative AI Hackathon 2025.
- Azure Hub-Spoke HA Platform** - Hub-spoke VNets with peering, NSG segmentation, and Bastion-only admin; Application Gateway (WAF) fronting a VM Scale Set behind a Standard Load Balancer.

ACHIEVEMENTS & HONOURS

- SEBI Securities Market Hackathon 2025 - Winner, Jury Recommendation Prize;** selected among the top 6 teams nationally.
- Google Generative AI Hackathon 2025 - Top 60 Nationwide** for an offline-first multilingual education PWA.
- Samsung Spatial Hackathon 2025 - 2nd Place;** ML dereverberation pipeline tuned to 48.2 GMAC/s under a 50 GMAC/s compute budget.
- SanDisk Hackathon (ECE Firmware)** - built CacheSense solo, an entropy-gated adaptive SSD compression pipeline benchmarked from 10 MB to 10 GB.
- Samsung PRISM - Selected Student Researcher** for Samsung's university research programme.
- Two published Indian patents as co-inventor** (App. Nos. 202641092083 and 202641062765).

TECHNICAL SKILLS

Networking: TCP/IP, OSI model, CIDR and subnetting, DNS, HTTP/HTTPS, WAN topology and device hierarchies, routing and link adjacency, fault correlation and incident triage, AQM and congestion control (CoDel, fq_codel), Linux tc/qdisc, eBPF

Cloud & Monitoring: Azure VNets, NSGs, VNet peering, Application Gateway, Load Balancer, Private Endpoints, Bastion, Function Apps; Kusto (KQL), Azure Monitor, Log Analytics, Network Watcher; Docker, Linux, Git, GitHub Actions, CI/CD

Programming & Data: Python, C++, Java, SQL, Bash, PowerShell, Verilog; Pandas, NumPy, scikit-learn, TensorFlow, model calibration, information retrieval and ranking, LLM integration and multi-agent workflows

Financial Technology: Securities market data and NSE/BSE APIs, market surveillance and credibility signals, regulatory compliance, privacy-preserving verification, risk scoring

Hardware: Cadence Virtuoso, Xilinx Vivado, Verilog RTL, FPGA, low-power VLSI

EDUCATION

Vellore Institute of Technology, Chennai | B.Tech, Electronics and Communication Engineering | CGPA 8.65/10 2023 - 2027

Indian Institute of Technology Madras | BS, Data Science and Applications 2024 - 2028

CERTIFICATIONS & LEADERSHIP

Certifications: IBM: Machine Learning with Python, Generative AI, Cloud Computing, Data Structures; Algorithmic Toolbox (Coursera) - verified badges on [Credly](https://www.credly.com). IEEE Photonics Society, VIT Chennai - Design Member (event graphics and creatives); BIS VIT Chapter; volunteer teacher with a community social-service initiative. Languages: English, Tamil, Hindi